Post-Decision Surprise in Charity Evaluation

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1 Introduction

This essay makes several comments on an ongoing discussion about how to best overcome the problem of post-decision surprise when deciding how to allocate donations to the most cost effective global health interventions.

I begin by summarizing the theoretical foundations of post-decision surprise, as coined by Smith and Winkler, and putting them in the context of charity evaluation.[1] Post-decision surprise is similar to the winner's curse in auction theory,[2] but applies to a single agent's estimates of the value of multiple assets, rather than multiple agents' estimates of the (common) value of a single asset.

In short, an agent presented with *unbiased*, ranked estimates of the expected cost effectiveness of multiple interventions will systematically overestimate the cost effectiveness of the top-ranked option. The expected magnitude of this error increases in the variance of the original estimate, which implies that even a risk neutral agent optimizing for cost effectiveness should not always choose to fund the intervention with the highest estimated cost effectiveness. As Smith and Winkler point out, there is a conceptually elegant solution to this problem, employing Bayes' theorem to completely eliminate post-decision surprise.

I then use recent quantified uncertainty analyses by Nolan, Rokebrand, and Rao (2022) to attempt Bayesian adjustments of GiveWell's actual cost effectiveness estimates.[3] In the absence of a clear

1. James E Smith and Robert L Winkler, "The optimizer's curse: Skepticism and postdecision surprise in decision analysis," *Management Science* 52, no. 3 (2006): 311–322.
2. Richard Thaler, *The Winners Curse: Paradoxes and Anolomies of Economic Life*, 1992.
3. Sam Nolan, Hannah Rokebrand, and Tanae Rao, *Quantifying Uncertainty in GiveWell CEAs*, Available at https://observablehq.com/@hazelfire/quantifying-uncertainty-in-givewell-ceas (2022/12/21).

method to systematically construct a prior distribution on true cost effectiveness, some plausible priors imply changes in the ranking of New Incentives, a charity providing households with small cash incentives for childhood vaccination. This is a cause for concern, as the ranking of charities on GiveWell's list potentially affects the allocation of millions of US dollars in annual donations.

Finally, I discuss my results, pointing out how some assumptions may be challenged. I also highlight a potential pitfall in the approach of only quantifying the uncertainty in several key parameters, which may otherwise seem attractive as a less labour-intensive method of comparing the uncertainty between cost effectiveness estimates.

Overall, identifying the precise difficulties in addressing post-decision surprise is the first step to solving them. These difficulties are still poorly understood, and extremely consequential.

2 Theoretical Foundations of Post-Decision Surprise

Consider a simple optimization problem. $\boldsymbol{\mu} = (\mu_1, ...\mu_n)$ is the true cost effectiveness of n interventions under consideration by a risk-neutral philanthropic funder. An analysis produces estimates $\mathbf{V} = (V_1, ...V_n)$, where $\mathbf{V}$ is an unbiased estimator of $\boldsymbol{\mu}$. We further assume that $\mathbf{V} = \boldsymbol{\mu} + \mathbf{e}$, where $\boldsymbol{e} = (e_1, ...e_n)$ and $e_i \sim N(0, \sigma_{e_i})$. The philanthropic funder then faces the following problem:

$$\underset{i\epsilon\{1,...,n\}}{\operatorname{argmax}} \left(E[\mu_i]\right)$$

Let i^* denote the intervention with the maximum V_i, that is: $\operatorname{argmax}_{i\epsilon\{1,...,n\}} V_i = i^*$. The **post-decision surprise** is the expected difference between the true cost effectiveness of i^*, and its estimated cost effectiveness: $E[\mu_{i^*} - V_{i^*}]$.

We have assumed that $\mathbf{V}$ is an unbiased estimator of $\boldsymbol{\mu}$, or equivalently, $E[\mathbf{V} - \boldsymbol{\mu}] = 0$. However, after we *select* for the intervention with the highest estimate, we should expect the post-decision surprise to be strictly negative for $n > 1$. In effect, selecting for the intervention with the highest estimate is selecting for both high true cost effectiveness μ_i, and high (positive) estimation error e_i. As a result, for $n > 1$, $E[\mu_{i^*} - V_{i^*}] > 0$; there is a systematic, foreseeable overestimation of the cost effectiveness of the top-ranked available intervention.

In some cases, post-decision surprise is not relevant to an agent's decision-making in this sim-

ple optimization problem. When the variance of the estimation error, σ_{e_i}, is equal for all i, i^* will also be the intervention with the highest expected true cost effectiveness: $\text{argmax}_{i\epsilon\{1,\ldots,n\}} V_i = \text{argmax}_{i\epsilon\{1,\ldots,n\}} E[\mu_i]$. However, there is no guarantee of this being the case when different interventions have different σ_{e_i}. In general, highly-ranked interventions with more uncertain estimates will also have post-decision surprise greater in magnitude than highly-ranked interventions with less uncertain estimates. This means that the agent's goal of choosing the intervention with the highest expected true cost effectiveness is not always best served by naïvely selecting the intervention with the highest estimate V_i.

Smith and Winkler (2006) argue that the solution to this problem is "conceptually quite simple".[4] If the uncertainty in the estimates is modelled explicitly, we can use Bayes' rule to determine the posterior distribution $\mathbf{v} = \boldsymbol{\mu}|\boldsymbol{V}$. The decision-maker then selects the intervention with the top-ranked posterior v. Supposing that i' is the solution to $\text{argmax}_{i\epsilon\{1,\ldots,n\}} v_i = i'$, the post-decision surprise is now:

$$\begin{aligned} E\big[\mu_{i'} - v_{i'}|\mathbf{V}\big] &= E\big[\mu_{i'} - E[\mu_{i'}|\mathbf{V}]|\mathbf{V}\big]. \\ &= E[\mu_{i'}|\mathbf{V}] - E[\mu_{i'}|\mathbf{V}] = 0 \end{aligned}$$

In other words, a funder selecting the intervention with the highest *posterior* estimated mean will not face the problem of post-decision surprise.

3 Bayesian Adjustments for Post-Decision Surprise in Practice

Nolan, Rokebrand, and Rao (2022) recently published a complete refactoring of GiveWell's cost effectiveness model, making it probabilistic. Instead of estimating cost effectiveness using point estimates of parameters, the 100-200 parameters for each intervention were converted to probability distributions. This was completed using either the confidence intervals in the constituent trials and meta-analyses, or–when such data was unavailable or unsuitable–arbitrary uncertainties in accordance with the authors' best guesses.

4. Smith and Winkler, "The optimizer's curse: Skepticism and postdecision surprise in decision analysis," 315.

There are strong reasons (as noted by the authors) for not taking these estimates literally. I use this data source in this essay because a) it is the only published uncertainty quantification of GiveWell's model, and b) it allows for a convenient illustration of methods that can be used for more rigorous models if and when they become available. As such, the figures and diagrams in this section should be taken as merely illustrative.

Smith and Winkler point out the following useful result.[5] Suppose (i) that the distributions of the true cost effectiveness of each intervention are independent and normal, such that $\mu_i \sim N(\bar{\mu}_i, \sigma^2_{\mu_i})$, and (ii) that the conditional distribution of estimated cost effectiveness is normal, such that $V_i|\mu_i \sim N(\mu_i, \sigma^2_{V_i})$. Then, the posterior distribution is $\mu_i|V_i \sim N(\hat{v}_i, \sigma^2_{\mu_i})$, where:

$$\textbf{Variance ratio: } \alpha_i = \frac{1}{1+\sigma^2_{V_i}/\sigma^2_{\mu_i}}$$
$$\textbf{Posterior mean: } \hat{v}_i = \alpha_i V_i + (1-\alpha_i)\bar{\mu}_i$$
$$\textbf{Posterior s.d.: } \sigma^2_{\mu_i|V_i} = (1-\alpha_i)\sigma^2_{\mu_i}$$

The variance ratio measures how much variance is in the prior distribution, as compared with the variance of the estimate. As the variance ratio increases, less weight is placed on the prior when calculating the posterior mean.

In order to apply this result to the available estimates, I first normalize them. Figure 1 demonstrates the distributions before and after normalization. To reiterate, two key assumptions of this analysis are that the estimates can be suitably approximated as normally distributed, and that they are independent (i.e., their covariance matrix is diagonal). Both of these assumptions are interrogated in the proceeding section of this essay.

5. Smith and Winkler, "The optimizer's curse: Skepticism and postdecision surprise in decision analysis," 317.

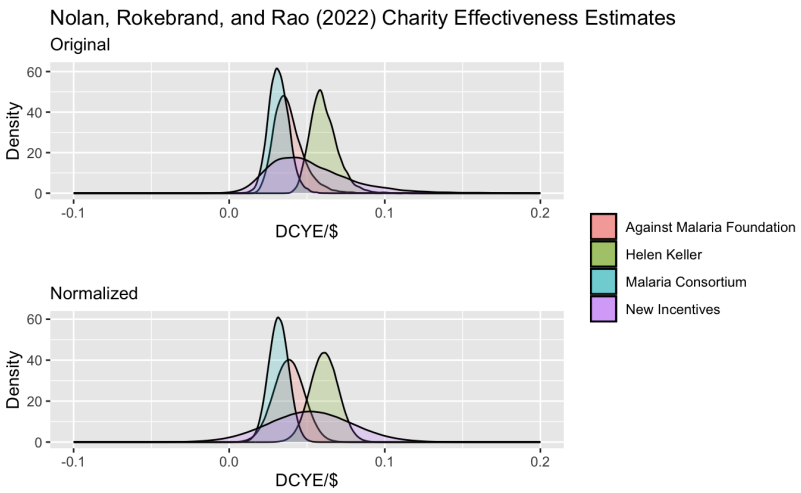

Figure 1: Normalized charity effectiveness estimates

Bayesian adjustment requires a setting a prior distribution, which is a difficult task. Analysts' beliefs about the effectiveness of GiveWell 'Top Charities' are likely inextricable from the research already conducted–even the y-axis units of Doubling Consumption-Year Equivalents (DCYE) would unlikely be used if GiveWell had not identified GiveDirectly (cash transfers to households in absolute poverty) as a promising intervention several years ago.[6] I present two plausible priors, one slightly above the cost-effectiveness of GiveDirectly (which is not a Top Charity, but included due to its common role as a benchmark), and one closer to the median of the four Top Charities. In the following figures, the grey area represents the prior distribution, ±1 standard deviation.

6. GiveWell, *GiveDirectly – November 2020 version*, Available at https://www.givewell.org/charities/give-directly/November-2020-version (2022/12/21).

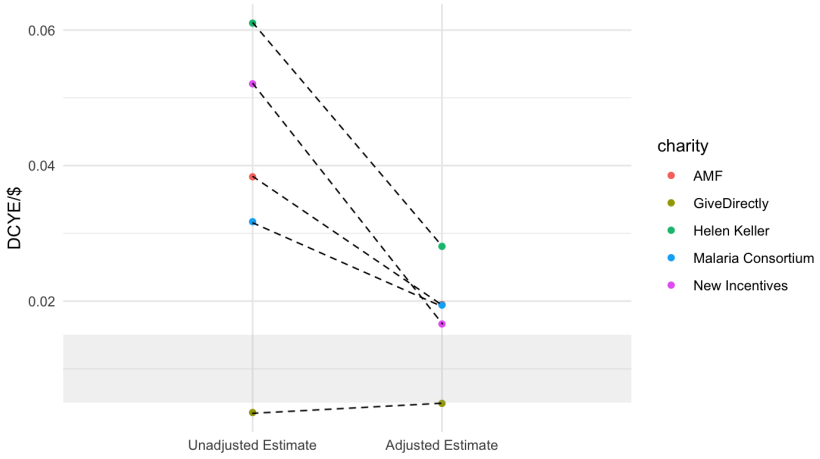

Figure 2: Bayesian adjustment with low prior

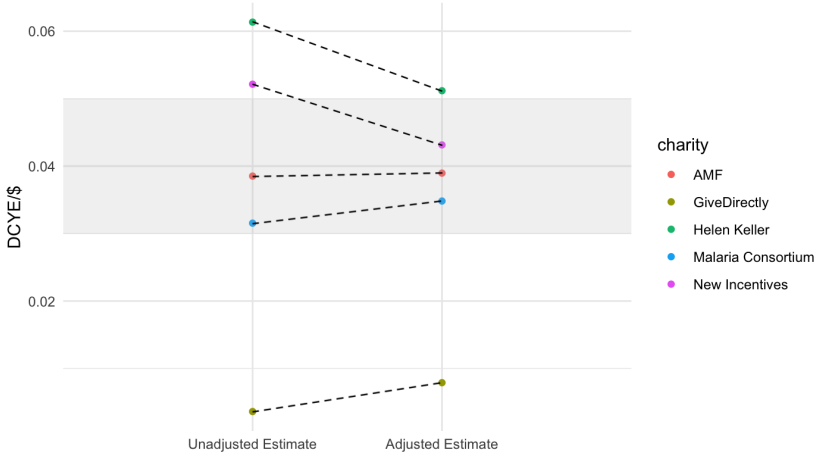

Figure 3: Bayesian adjustment with high prior

Figure 2 shows that one charity, New Incentives, declines from GiveWell's second-ranked charity to its fourth-ranked charity after Bayesian adjustment with a lower prior. In contrast, Figure 3, with a higher prior, shows no change in overall resulting from the adjustment. If explicit Bayesian adjustment is to be used as a method of overcoming post-decision surprise, future researchers will have to find systematic, transparent ways of generating a prior distribution on charity cost effective-

ness. Perhaps, this will come as a result of closer examination of GiveWell's early research, such as counting how many charities have been considered, or aggregating the results of all cost effectiveness estimates ever published by GiveWell.

4 Discussion

The theory in Smith and Winkler (2006) tells us that post-decision surprise is a problem that charity evaluators should not ignore, and that using posterior distributions calculated using Bayes' rule is its solution. In practice, however, this only gets us part of the way. In order to employ explicit Bayesian adjustments, we must identify a reasonable, and defensible prior on the cost effectiveness of evaluated charities.

Additionally, putting the issue of constructing a prior distribution aside, a more rigorous Bayesian adjustment would take into account (at least) two additional complicating factors:

First, the most sensible prior distributions $\boldsymbol{\mu}$, as well as the cost effectiveness estimates $\mathbf{V}$, may not be distributed in such a way that makes them appropriate to approximate with a distribution amenable to simple applications of Bayes' rule (e.g., normal distributions, beta distributions). My previous assumption was that the prior distribution, and GiveWell's estimates were normally distributed. However, a) it is plausible that charity cost effectiveness is better approximated as lognormal (with most charities doing close to $0\times$ cash transfer equivalents, and some doing $10\times$ or even $100\times$ cash transfer equivalents), and b) Nolan, Rokebrand and Rao's estimates are probably not best approximated as normal distributions (see Figure 1).

Second, cost effectiveness estimates of different interventions often share parameters (e.g., the population in a country with a certain disease). My previous assumption was that GiveWell's estimates were conditionally independent given $\boldsymbol{\mu}$, but–since they share parameters–this is evidently not the case. A more rigorous Bayesian adjustment would use the actual (non-diagonal) covariance matrices.

These two technical problems are already thoroughly discussed in the health economics and decision science literature, such as in Briggs (2006).[7] They should not preclude future research

7. Andrew Briggs, Mark Sculpher, and Karl Claxton, *Decision modelling for health economic evaluation* (Oxford University Press, 2006).

from developing rigorous Bayesian adjustments of charity cost effectiveness evaluations. As for the construction of a prior distribution, any solution will likely be somewhat subjective, and a lightning rod for criticism. However, formally expressing a prior is not necessarily worse than the post hoc adjustments GiveWell currently uses, such as a significant downwards adjustment on the cost effectiveness of deworming to account for the potential that the cited RCTs are not internally valid.[8] Whether it uses post hoc adjustments, develops an explicit prior, or does nothing at all, GiveWell's analysis is inescapably contingent on *some* prior belief about charities' cost effectiveness.

All adjustments for post-decision surprise hinge on the observation that value estimates often come with different levels of uncertainty. This makes comparing the uncertainties between estimates necessary to address this problem. However, GiveWell seems reluctant to explicitly quantify uncertainty in each parameter of their model, as attempted in the estimates discussed above. In a recent note, Isabel Arjmand, GiveWell Special Projects Officer, cites time-intensiveness and reduced legibility as two considerations against complete uncertainty quantification, but notes that evidence of a "sufficiently large impact on the value of [GiveWell's] allocations" may outweigh these costs.[9]

One proposed compromise is to conduct a probabilistic sensitivity analysis (PSA), which "would involve selecting and parameterizing probability distributions for key parameters; running repeat simulations to obtain a distribution of potential outcomes; and using this distribution as the basis for decision-making."[10] This is not necessarily a bad solution, but charity evaluators must be careful in not being overly restrictive in which "key" parameters they select. I will illustrate a potential issue that could arise through a stylized example.

Analysis of Charity A: This analysis relies on some very promising trials for the efficacy of the intervention Charity A uses. The trials have small confidence intervals, but took place in different contexts (e.g., a different city) with slightly different methods (e.g., researchers checked up on patients more often in the trial) than those used by Charity A. As such, the cost effectiveness analysis includes numerous parameters roughly adjusting the trial's results for potential internal and external validity concerns.

8. GiveWell, *Combination Deworming (Mass Drug Administration Targeting Both Schistosomiasis and Soil-Transmitted Helminths)*, Available at https://www.givewell.org/international/technical/programs/deworming (2022/08/01).

9. Isabel Arjmand, *The winners of the Change Our Mind Contest—and some reflections*, Available at https://blog.givewell.org/2022/12/15/change-our-mind-contest-winners/ (2022/12/16).

10. Arjmand.

Analysis of Charity B: This analysis relies on some very promising trials for the efficacy of the intervention Charity B uses. The trials have wider confidence intervals, but take place in a very similar context, using identical methods to Charity B.

Tn this example, if the "key" parameters selected by the charity evaluator were exclusively those with an easily defensible, objective means by which to quantify uncertainty (i.e., the confidence intervals from the trials), then Charity B might unjustifiably have a higher estimated uncertainty than Charity A. This is because the charity evaluator's decision to focus on only some categories of parameters is punitive towards charities with higher uncertainty in those categories, and forgiving towards charities whose parameters may be highly uncertain in other categories (e.g., due to external validity or something else that escapes scientific study).

To conclude, post-decision surprise poses a difficult challenge to charity evaluators and the philanthropic funders which rely on their research. As I have endeavoured to show in this essay, a solution to this problem is unlikely to be straightforward, and will deserve further attention from health economists, decision scientists, and statisticians.

References

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